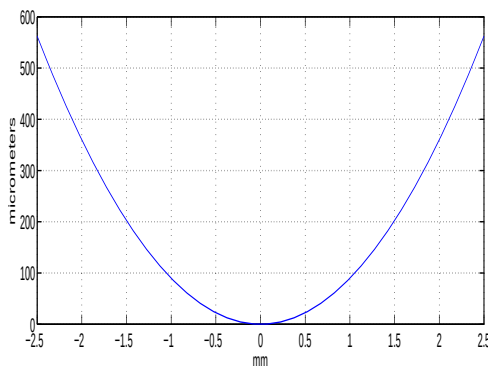


Math 1452: Interesting application problems for Sections 7.3 and 7.8

- (1) The upper meniscus of a water drop in a pipette (see Figure 1(a)) may be modeled as a surface obtained by revolving the function $y = 90x^2$ about the y -axis, where x is in millimeters and y is in micrometers. Find the surface area of the meniscus.



(a) A pipette with water. The upper meniscus shows reflections from the given by $y = 90x^2$, where x is in millimeters, and y is in micrometers.

Figure 1: Surface area of a meniscus in a capillary tube.

Solution: As the units on y and x are different, we need to re-scale y so that the units are the same: $y = ax^2$ millimeters, where $a = 9 \times 10^{-2}$. Let $R = 2.5$ mm.

The surface area is given by:

$$\text{Surface area} = \int_0^R 2\pi x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = 2\pi \int_0^R x \sqrt{1 + 4a^2 x^2} dx.$$

Use the substitution: $2ax = \tan(\theta)$. Then: $\sqrt{1 + 4a^2 x^2} = \sec(\theta)$ and $dx = \frac{1}{2a} \sec^2(\theta) d\theta$.

$$\int x \sqrt{1 + 4a^2 x^2} dx = \frac{1}{4a^2} \int \tan(\theta) \sec(\theta) \sec^2(\theta) d\theta.$$

Set $du = \tan(\theta) \sec(\theta) d\theta$. Then $u = \sec(\theta)$ and

$$\int x \sqrt{1 + 4a^2 x^2} dx = \frac{1}{4a^2} \int u^2 du = \frac{1}{4a^2} \frac{u^3}{3} + C = \frac{1}{4a^2} \frac{\sec^3(\theta)}{12a^2} + C = \frac{(1 + 4a^2 x^2)^{\frac{3}{2}}}{12a^2} + C.$$

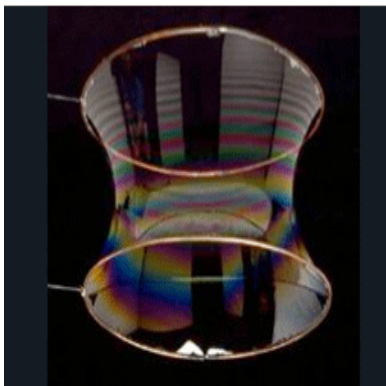
Therefore,

$$\text{Surface area} = 2\pi \frac{(1 + 4a^2 x^2)^{\frac{3}{2}}}{12a^2} \Big|_0^R = 2\pi \frac{(1 + 4a^2 R^2)^{\frac{3}{2}} - 1}{12a^2} = 20.6 \text{ mm}^2.$$

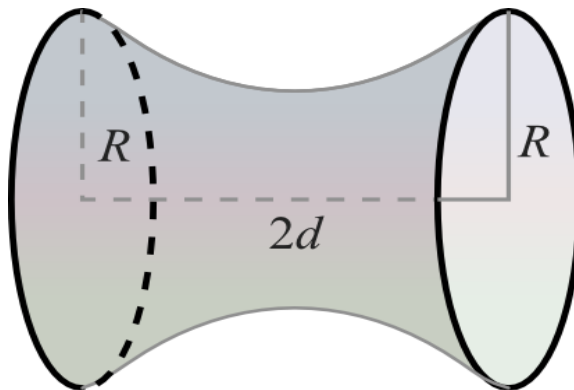
Note that the area of the circular cross-section of the pipette $= \pi a^2 = 19.63 \text{ mm}^2$, which means that the meniscus surface is about 5% larger than the cross-section.

□

- (2) A soap film between two circles at a distance $2d$ from each other (see Figure 2(a)) may be modeled as a surface obtained by revolving the function $y = \cosh(x) = \frac{e^x + e^{-x}}{2}$ about the x -axis. Find the surface area of the film.



(a) A soap film between two circles.



(b) After appropriate scaling, the film surface may be modeled as a surface obtained by revolving $y = \cosh(x)$ about the x -axis.

Figure 2: Surface area of a soap film between two circles. The shape of the film is called a *catenoid*.

Solution:

The surface area is given by:

$$\begin{aligned}
 \text{Surface area} &= \int_{-d}^d 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \\
 &= 2\pi \int_{-d}^d y \sqrt{1 + \sinh^2(x)} dx \\
 &= 2\pi \int_{-d}^d \cosh^2(x) dx \\
 &= \pi \int_{-d}^d (1 + \cosh(2x)) dx \\
 &= \pi \left(x + \frac{\sinh(2x)}{2} \right) \Big|_{-d}^d \\
 &= 2\pi \left(d + \frac{\sinh(2d)}{2} \right).
 \end{aligned}$$

□